

# Load Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q19-1312

Juniata 138 kV  
October 25, 2020

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1312. The Interconnection Customer proposes a 135.3 MW large load (data center) facility interconnecting at the Juniata 138 kV substation in the Pennsylvania area. The request was received on October 14, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 493 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$81,719,000.

### 3. Network Impacts

#### 3.1 Load Deliverability Analysis

| Monitored Facility                      | Contingency                                                    | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------|----------------------------------------------------------------|-------------|--------------|----------|
| LITTLE CHUTE 345/138kV Transformer Bank | LITTLE CHUTE – TALENENERGY MARTINS CREEK LLC HARRISBURG 345 KV | 141.0       | 382          | 10.5     |
| ELK CITY LLC – JOHN HART 138 KV line    | LITTLE CHUTE – TALENENERGY MARTINS CREEK LLC HARRISBURG 345 KV | 132.7       | 293          | 33.5     |
| USAA 2 – CORONET 138 KV                 | FT HUMBUG – LITTLE CHUTE 345 KV                                | 122.5       | 320          | 31.2     |
| CORONET 345-138 kV Autotransformer      | CORONET – WESTWOOD GENERATION LLC 500 KV                       | 119.9       | 522          | 26.8     |
| PINGSTON – CALAVERAS 138 KV             | ELK CITY LLC – JOHN HART 138 KV                                | 119.4       | 204          | 10.1     |
| CALAVERAS – PANGUITCH 138 KV            | ELK CITY LLC – JOHN HART 138 KV Line                           | 117.8       | 228          | 10.1     |
| USAA 2 – GARLAND 138 KV                 | ELK CITY LLC 138 KV/1 kV Auto                                  | 117.0       | 251          | 47.7     |
| PINGSTON – JACK CRK 138 KV              | ELK CITY LLC – JOHN HART 138 KV                                | 105.6       | 207          | 10.1     |
| PANGUITCH – ELK CITY LLC 138 KV Line    | JOHN HART – ELK CITY LLC 138 KV                                | 103.4       | 279          | 10.1     |
| EKLUTA – ALLEGHENY HYDRO NO 8 345 KV    | CALDWELL – PINEY CREEK PROJECT 345 KV Line                     | 100.9       | 1,188        | 5.5      |
| CORONET 500 KV/345 KV XFMR              | WESTWOOD GENERATION LLC – CORONET 500 KV                       | 100.1       | 939          | 11.0     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------|---------------------------------------------|---------------|-----------------------|
| LITTLE CHUTE 345/138kV Transformer Bank      | Replace transformer with higher-rated unit  | \$17,333,000  | 100% (\$17,333,000)   |
| JOHN HART – ELK CITY LLC 138 KV              | Rebuild line with higher-capacity conductor | \$726,000     | 100% (\$726,000)      |
| USAA 2 – CORONET 138 KV                      | Rebuild line with higher-capacity conductor | \$5,273,000   | 100% (\$5,273,000)    |
| CORONET 345-138 kV Autotransformer           | Replace transformer with higher-rated unit  | \$15,569,000  | 100% (\$15,569,000)   |
| PINGSTON – CALAVERAS 138 KV                  | Rebuild line with higher-capacity conductor | \$5,340,000   | 100% (\$5,340,000)    |
| CALAVERAS – PANGUITCH 138 KV                 | Rebuild line with higher-capacity conductor | \$3,117,000   | 100% (\$3,117,000)    |
| USAA 2 – GARLAND 138 KV                      | Rebuild line with higher-capacity conductor | \$6,479,000   | 100% (\$6,479,000)    |
| PINGSTON – JACK CREEK SUBSTATION 138 KV      | Reconductor line                            | \$2,231,000   | 100% (\$2,231,000)    |
| PANGUITCH – ELK CITY LLC SUBSTATION 138 KV L | Reconductor line                            | \$872,000     | 100% (\$872,000)      |
| EKLUTNA – ALLEGHENY HYDRO NO 8 345 KV line   | Reconductor line                            | \$24,263,000  | 4% (\$984,000)        |
| CORONET 500-345 kV Autotransformer           | Replace transformer with higher-rated unit  | \$23,795,000  | 100% (\$23,795,000)   |
|                                              | Total allocated to project                  |               | \$81,719,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.